

C# BACKEND TRACK | OOP PHASE

Task 1

# System Analysis Task

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Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

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## SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

|                         |                         |                                                                                                                                                              |
|-------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Generated</b> | <b>System Generated</b> | The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.                          |
| <b>User Input</b>       | <b>User Input</b>       | The user types this value directly at the keyboard when prompted by the system.                                                                              |
| <b>Default Value</b>    | <b>Default Value</b>    | A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".                           |
| <b>Calculated</b>       | <b>Calculated</b>       | Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor. |
| <b>From List</b>        | <b>From List</b>        | The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.                           |

## Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

### Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

| Entity Name: <b>Book</b> |           |                  |
|--------------------------|-----------|------------------|
| Property Name            | Data Type | Source Tag       |
| BID                      | string    | System Generated |
| title                    | string    | User Input       |
| author                   | string    | User Input       |
| genre                    | string    | From List        |
| copiesInStock            | int       | Default Value    |
| isAvailable              | bool      | Calculated       |
|                          |           |                  |

| Entity Name: <b>Member</b> |           |                  |
|----------------------------|-----------|------------------|
| Property Name              | Data Type | Source Tag       |
| MID                        | string    | System Generated |
| name                       | string    | User Input       |
| dateOfBirth                | date      | User Input       |
| phoneNumber                | string    | User Input       |
| email                      | string    | User Input       |
|                            |           |                  |
|                            |           |                  |

| Entity Name: Librarian |           |                    |
|------------------------|-----------|--------------------|
| Property Name          | Data Type | Source Tag         |
| LID                    | string    | Computer Generated |
| name                   | string    | User Input         |
| phoneNumber            | string    | User Input         |
| email                  | string    | User Input         |
| role                   | string    | Default Value      |
|                        |           |                    |
|                        |           |                    |

| Entity Name: BorrowedBook |           |                 |
|---------------------------|-----------|-----------------|
| Property Name             | Data Type | Source Tag      |
| BBID                      | string    | SystemGenerated |
| member                    | string    | Calculated      |
| book                      | string    | Calculated      |
| MaxDays                   | int       | Default Value   |
| borrowDate                | date      | Default Value   |
| overdueDays               | int       | Calculated      |
| fine                      | decimal   | Calculated      |
| status                    | string    | Default Value   |

| Entity Name: <b>Reservation</b> _____ |           |                  |
|---------------------------------------|-----------|------------------|
| Property Name                         | Data Type | Source Tag       |
| RID                                   | string    | System Generated |
| member                                | string    | Calculated       |
| book                                  | string    | Calculated       |
| reservationDate                       | date      | Default Value    |
| status                                | string    | Default Value    |
| isNotified                            | bool      | Default Value    |
| _____                                 | _____     | _____            |

*If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.*

## Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

The system allows a member to register (Member entity created) and/or librarian (Librarian entity) to login. A book can be added to the library (Book entity created). Member can borrow a book only if the book is available in stock (new BorrowedBook entity created) and the Book entity's copiesInStock will get decreased in order, set borrowDate and status. Then, if there are no more copies (copiesInStock == 0); isAvailable property get switched to false. In addition, a book can be reserved by a member (create new Reservation) and will get notified once the book is available.

**Step 3 — List the Services (Phase 4)**

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

| #  | Service Name        | What it does                                                  |
|----|---------------------|---------------------------------------------------------------|
| 1  | RegisterNewMember   | Create new Member entity                                      |
| 2  | AddBook             | Add new book to the library                                   |
| 3  | AddCopy             | Add copy of an existing book to the library                   |
| 4  | DeleteBook          | Delete a book from the library system (or set as unavailable) |
| 5  | UpdateMember        | Update member information                                     |
| 6  | DeleteMember        | Delete member from the system                                 |
| 7  | BorrowBook          | Add BorrowedBook entity; Assign all details                   |
| 8  | ReserveBook         | Add Reservation entity; assign all details                    |
| 9  | ViewAllborrows      | Display all BorrowedBook entities with details                |
| 10 | ViewAllReservations | Display all Reservation entities with details                 |

## Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

**Your task:** For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

| Guest                         |                     |                         |
|-------------------------------|---------------------|-------------------------|
| Property Name                 | Data Type           | Source Tag (fill in)    |
| <code>guestId</code>          | <code>int</code>    | <u>System Generated</u> |
| <code>guestName</code>        | <code>string</code> | <u>User Input</u>       |
| <code>guestPhone</code>       | <code>string</code> | <u>User Input</u>       |
| <code>guestEmail</code>       | <code>string</code> | <u>User Input</u>       |
| <code>guestNationality</code> | <code>string</code> | <u>User Input</u>       |
| <code>registrationDate</code> | <code>string</code> | <u>System Generated</u> |

| Room          |           |                      |
|---------------|-----------|----------------------|
| Property Name | Data Type | Source Tag (fill in) |
| roomId        | int       | System Generated     |
| roomNumber    | string    | User Input           |
| roomType      | string    | From List            |
| floorNumber   | int       | Calculated           |
| nightlyPrice  | decimal   | Calculated           |
| isAvailable   | bool      | Default Value        |

| Booking        |           |                      |
|----------------|-----------|----------------------|
| Property Name  | Data Type | Source Tag (fill in) |
| bookingId      | int       | System Generated     |
| guestId        | int       | From List            |
| roomId         | int       | From List            |
| checkInDate    | string    | User Input           |
| checkOutDate   | string    | User Input           |
| numberOfNights | int       | Calculated           |
| totalPrice     | decimal   | Calculated           |
| status         | string    | Default Value        |



| CheckInRecord               |           |                      |
|-----------------------------|-----------|----------------------|
| Property Name               | Data Type | Source Tag (fill in) |
| <code>recordId</code>       | int       | System Generated     |
| <code>bookingId</code>      | int       | Calculated           |
| <code>guestId</code>        | int       | Calculated           |
| <code>roomId</code>         | int       | Calculated           |
| <code>actualCheckIn</code>  | string    | System Generated     |
| <code>actualCheckOut</code> | string    | System Generated     |
| <code>finalAmount</code>    | decimal   | Calculated           |

## Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

In the system, a guest can register himself (new Guest entity), and book a room with entering required fields (new Booking record, calculate totalPrice from numberOfNights and room's nightlyPrice, and update room isAvailable field). The guest may check-in which will create a (CheckInRecord which gets bookingId, guestId, and roomId from related objects) or he can cancel his booking (update room isAvailable field). The system admin should be allowed to add new rooms.

## Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

| # | Service Name  | What it does                         |
|---|---------------|--------------------------------------|
| 1 | AddNewRoom    | Create new Room                      |
| 2 | DeleteRoom    | Delete Room (or make it unavailable) |
| 3 | UpdateRoom    | Update Room details                  |
| 4 | ViewAllRooms  | Display all Rooms with their details |
| 5 | RegisterGuest | Create new Guest                     |
| 6 | UpdateGuest   | Update Guest information             |
| 7 | BookRoom      | Create new Booking                   |

| # | Service Name  | What it does                       |
|---|---------------|------------------------------------|
| 8 | CheckIn       | Create new CheckInRecord           |
| 9 | CancelBooking | Update Booking status to Cancelled |